



Original article

The cross-level impact of landscape patterns on housing premiums in micro-neighborhoods

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ABSTRACT

This paper attempts to examine the impacts of the spatial structure of landscape on housing premiums in Austin, TX through the years of 2009–2011. Dissimilar to previous studies, this paper highlighted three points. First, landscape patterns was observed—not only the composition but also the configuration in terms of shape, fragmentation, isolation, and connectivity. Second, the definition of neighborhood in this study was closely matched to the nature of a residential neighborhood, which was represented by a residential subdivision that has a name and a high level of homogeneous characteristics (i.e., development age, appraised value, lot size, and housing size). This approach specifically helps understand the impact of landscape configuration at the micro-level neighborhoods. Third, the hierarchical nature of neighborhoods and embedded parcels—nested parcels within the same neighborhood posit a certain level of identity as a whole—was given attention so that multi-level modeling could be employed. Conceptually and theoretically, multi-level modeling is a better approach to examine phenomenon that occur in multi-level units that show a clear hierarchy. The findings indicated that home buyers are willing to pay more to live in neighborhoods with the high ratio of tree cover. This is a finding consistent with previous studies. Meanwhile, the detailed spatial configuration of landscape does not play an important role at the residential neighborhood level. This result urges policy makers to be more scale-sensitive when planning landscape and indicates that micro neighborhoods are not the correct spatial level to discuss the spatial configuration of landscape, greenways, and green network.

1. Introduction

One aim of community design and planning is to encourage the construction of high quality landscapes because human beings are an vital element of landscape who are legitimate receivers of benefits that could be brought by landscape. On the contrary, sometimes poor community planning can lead to disturbances of the landscape ecology. Previous landscape theories support the idea that landscape can provide restorative, recreational, and esthetic benefits to people and also promote the place attachment and place identity that can induce behavior commitments to surrounding settings (Hunziker et al., 2007; Lewicka, 2011, p. 219). Previous empirical studies also reported positive contributions of human perception toward landscape for stress relief (Kaplan, 2001; Ulrich et al., 1991), neighborhood satisfaction (Lee et al., 2008), housing premiums (Jim and Chen, 2009; Rodriguez and Sirmans, 1994), physical, social, and educational benefits (Donovan et al., 2002; Humpel et al., 2004, 2002; Kloek et al., 2013), and promoted place attachment and behavior commitments (Hernandez et al., 2007; Lewicka, 2011; Scannell and Gifford, 2010).

Also, researchers found that landscapes, functioning as critical elements of other infrastructure, can mediate flood damage (Brody et al., 2013; Gill et al., 2007; Kim and Park, 2016) and urban heat island effects (Connors et al., 2013; Liu and Weng, 2008), as well as protect air quality and natural habitats (Benedict and McMahon, 2006; McDonald et al., 2005). It is also well known that various forms of landscape could affect neighborhood satisfaction that measures overall quality of well-being reflected in both stated and revealed preferences (Des Rosiers et al., 2002; Kweon et al., 2010; Lee et al., 2008; Stigaril and Elam, 2009; Vemuri et al., 2009).

Even though previous research is rich in finding associations between human response to landscape, there are several points that have not been fully tested. Landscape ecology theory emphasized the importance of roles of both composition and configuration of landscape on human and other ecosystems (Ndubisi, 2002). But, most empirical studies focused heavily on the types, proximity, quality, and amount of individual elements of landscape. Few empirical studies explored the spatial structure and landscape ecology indices to identify the role of landscape structure on perception and evaluation of people reflected in

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neighborhood satisfaction, which measure the overall well-being (Yang, 2008). Lee et al. (2008) tested the impact of landscape structure and neighborhood satisfaction to identify the evaluation of people, but their data were collected by survey. Although surveys can be a meaningful way to measure neighborhood satisfaction, typically their sample size should be small. Given the facts, this paper attempts to observe the impact of both of composition and configurations of landscape on neighborhood satisfaction as reflected in 9958 parcels nested in 649 subdivisions in Austin, TX from 2009 to 2011. There are relatively larger samples compared to revealed preference studies. In addition, this association was tested through multi-level linear modeling to provide conceptually and statistically more appropriate explanations of the hierarchical nature of neighborhoods and nested parcels. The results of this study are expected to serve as a foundation of landscape planning, landscape management, and land use decisions in practices that utilize more spatially and statistically sensitive approaches.

2. Literature review

2.1. Landscape and human relationship

Human beings are an essential part of landscape ecology (Naveh, 2000); therefore, landscape ecology research has highlighted the importance of understanding the interactions between human and environment relationships (Turner et al., 2001). According to Hunziker et al. (2007), the human-landscape relationship could be categorized into two ways; as physical “space” and cultural and social “place” (Hunziker et al., 2007; Kienast et al., 2012; Ode and Miller, 2011). The first highlighted the role of landscape on restoration from stress and attentional fatigue (Kaplan, 2001; Kaplan, 1995; Ulrich et al., 1991), promotion of esthetical pleasure (Ulrich, 1983), and some other direct benefits such as emotional (e.g., escape from urban settings, feeling of freedom, and identifying with nature), intellectual (e.g., learning nature and local history), social (e.g., social gathering and team building), and physical aspects (e.g., outdoor recreation and feeling being in fresh air; Mostyn, 1979). Numerous empirical studies found the positive impacts of landscape on the promotion of physical activities, social activities, and relief from urban settings (Hayward and Weitzer, 1984; Jim and Chen, 2006; Özgüner and Kendle, 2006; Parsons et al., 1994). The latter regarded landscape as a critical element of increasing place attachment, place identity, and place dependence, each of which imply a positive bond between people and their environments, conscious and unconscious preference, feelings, values, and behavior tendencies, the strength of connections between occupants and specific places respectively (Jorgensen and Stedman, 2001). Empirical studies also found that people develop attachment, identification, and behavior commitment to their surrounding landscapes (Hartig et al., 1997; Hernandez et al., 2007; Rishbeth and Powell, 2013) and a promoted sense of place often becomes strong motivation for community actions including landscape conservation practices in turns (Erickson et al., 2002; Theodori, 2004).

2.2. Landscape patterns on housing premium

Even though it is clear that the association between landscapes and humans should be observed into two modes—sense of space and sense of place—, neighborhood satisfaction studies often combine these two concepts since neighborhood satisfaction arises from the subjective and objective evaluations of occupants experience in neighborhoods (Amérigo and Aragones, 1997). Theoretically, direct and objective benefits from landscape such as economic, recreational, aesthetic, and restorative benefits is likely to increase neighborhood satisfaction (Lee et al., 2008) and therefore promoted place attachment, identity, and dependence (French et al., 2013). Because of the limited data availability of sense of place that required detailed survey data, numerous studies used housing price as a useful proxy to quantitatively measure

neighborhood satisfaction that encompasses the concept of sense of place by focusing on the characteristics of “space.”

In academic research, neighborhood satisfaction is typically measured by two types of preferences: state and revealed. The stated preference approach primarily relies on surveys that evaluate the satisfaction or quality of a neighborhood as a whole or based on specific conditions. Alternatively, the revealed preference approach focuses on market prices that were paid for properties and assumes that people are willing to pay for the neighborhood characteristics they prefer. Both have pros and cons, but it is clear that the observation of housing prices serves as a powerful means to evaluate the quality of neighborhood settings. Previous studies have examined the associations between different types of landscape features and levels of neighborhood satisfaction as revealed by housing premiums. Particularly, the existence, amount, view, or distance toward landscape elements including urban parks, trees, water features, or several types of open space has been identified to have a positive association with housing (or apartment) transaction prices (Baranzini et al., 2010; Baranzini and Schaefer, 2011; Jim and Chen, 2010; Luttik, 2000; Mansfield et al., 2005; Sander et al., 2010; Schläpfer et al., 2015; Wen et al., 2014, 2015; Yoo and Wagner, 2016).

As shown above, the impacts of composition of landscape on human perception and activities on neighborhood satisfaction have been widely studied, while the configuration has been relatively less tested, though landscape ecologists argue that both configuration and composition are critical factors defining the human-landscape relationship (Ndubisi, 2002). If the composition and configuration of landscape are critical circumstances in human dimension, how can we define the better or worse condition? There is no absolute number and standard, yet some principles were commonly adopted in empirical studies. One of the most popular spatial guidelines, proposed by Forman (1995) and Shafer (1994), assume that heterogeneous sizes of patches, wide corridors, less fragmented and isolated, and more connected landscape could serve better for the ecosystem because these conditions could facilitate flow and movement of species and prevent habitat loss (Lee et al., 2008). Based on these assumption and landscape ecology metrics, landscape planning and management tools utilized the concept to analyze the best area for creating greenways and habitat. Academic literature tested how the configuration affects the citizens' scenic beauty estimation (Antrop and Van Eetvelde, 2000), water quality (Jones et al., 2001), and biodiversity (Uuemaa et al., 2009). Lee et al. (2008) conducted a similar study in which we are interested, which found that less fragmented, less isolated, and well connected landscape would increase neighborhood satisfaction as measured by revealed preference. Given the fact, hypothetically, the different landscape mosaic would influence the level of neighborhood satisfaction and be reflected in housing prices.

2.3. Definition of neighborhoods and operationalization concerns

Even though researchers have adopted the term “neighborhood” as their unit of analysis, the units of data collection used to represent a neighborhood are by no means uniform. The most frequent neighborhood operation was to draw buffers by Euclidean or network distances, typically a quarter-mile, a half-mile, or one mile. A quarter to a half-mile corresponded to a 5-min walking distance, and one mile corresponded to the maximum acceptable walking distance (Lawhon, 2009). Buffer-drawn neighborhoods underscore the importance of personal familiarity and accessibility to one's immediate surroundings (Park and Rogers, 2015). Buffer-drawn neighborhoods prove to be efficient for data collection but not for fulfilling the general definition of neighborhoods, since each parcel contains its own neighborhood. Parcels within a neighborhood are typically bound by a soft boundary (e.g., subdivisions, gated communities, apartment complexes, fee-based communities, homeowners' associations, zip codes, or assessor's codes) or a hard boundary (e.g., administrative districts, geographic barriers:

ivers, streams, or roads) (Clapp and Wang, 2006). Additionally, each neighborhood carries a certain level of shared identity distinct from other neighborhoods. In particular, residential neighborhoods are characterized by similar designs in terms of landscape patterns, architectural style, street design, and socioeconomic status. They often share names and have property or homeowners associations that manage the neighborhood under covenants or deed restrictions (Park and Rogers, 2015). These types of neighborhoods—henceforth referred to as “real neighborhoods” to distinguish them from buffer-drawn neighborhoods—align more closely to the conceptual nature of neighborhoods.

Neighborhoods with predefined boundaries, however, may generate statistical issues when conducting ordinary least squares regression (OLS), since OLS assumes that every parcel within a neighborhood boundary shares identical characteristics while maintaining individual parcel conditions, resulting in a mixture of conditions from different spatial levels. For example, if Neighborhood A contains 20 percent tree cover, all nested parcels would have the same value of 20 percent, though each parcel's housing characteristics are not aggregated into the neighborhood level as shown in Eq. (1). Even past studies with a buffer-drawn neighborhood for each parcel often assigned socio-demographic or economic conditions retrieved from a census to nested parcels, in order to control for conditions other than the main interests of the studies. This approach, though commonly accepted, violates OLS because each individual observation is not independent of the others (Wooldridge, 2012). Thus, OLS no longer serves as an optimal means of estimation, and as a result, the standard errors of the effects of the explanatory variables and the dependent variable must be adjusted (Wech and Heck, 2004).

$$\text{Price}_{\text{Parcel}} = \beta_0 + \sum_{\text{Parcel}} \beta_{ij} X_{ij} + \sum_{\text{Neighborhood}} \beta_j X_j + \dots + \epsilon \quad \text{.....Spatial Levels} \quad (1)$$

where P signifies housing price; X_i represents the characteristics of parcel i in a neighborhood j ; X_j constitutes the neighborhood characteristics of j where parcel i is nested.

To avoid this statistical violation, the characteristics of the parcels can be aggregated up to neighborhood levels as described in Eq. (2). This approach, however, discards important information about the parcels; for example, using the mean or median housing values of each neighborhood dramatically reduces the sample size.

$$\text{Price}_m = \beta_0 + \sum_{\text{Neighborhood}} \beta_{mij} X_{mij} + \sum_{\text{Neighborhood}} \beta_j X_j + \dots + \epsilon \quad \text{.....Spatial Levels} \quad (2)$$

where Price_m signifies the mean or median housing price by each neighborhood; X_{mij} represents the mean or median value of parcel characteristics in a neighborhood j ; X_j constitutes the neighborhood characteristics of j where parcel i is nested.

To simultaneously accommodate for both parcel data and the conceptual nature of neighborhoods with a shared boundary, one can consider using multi-level modeling (MLM). MLM utilizes an ordinary linear squares (OLS) framework that includes two different stages. First, regressions are conducted separately among lower-level units (parcels in this study) within the same higher-level unit (subdivisions in this study). For example, if parcels “a” and “b” are nested in subdivision A and parcels “c” and “d” in subdivision B, two OLSs are conducted separately between parcels “a and b” and “c and d.” Second, the intercepts and slopes from these equations are averaged across the higher-level units and then evaluated based on the standard error of each estimate (Arnold, 1992). This process provides theoretical and empirical benefits through the following: (1) a simultaneous examination of relationships within and across levels by partitioning variance outcomes among lower and higher levels, without wasting information in lower order units and adjusting the standard errors of the relationships; (2) an adherence to the independency assumption of OLS (Poston, 2002; Raudenbush et al., 2011); and (3) an optimal theoretical

approach that addresses multi-level units.

A few previous studies were inspired by the richness of multi-level approaches to identify the association between housing price and new urbanism design by Yang (2008), transit-oriented development by Duncan (2010), neighborhood design features by Shin et al. (2011) operationalized by parcels and neighborhoods, 50- and 400-meter radius buffers and block groups, and parcels and subdivisions respectively. Though Yang (2008) and Shin et al. (2011) considered the composition of landscape, there was a lack of testing on the composition and configuration of landscape on housing prices based on a multi-level approach. Moreover, Lee et al. (2008) were aware that there might be difference of the spatial scales of neighborhoods so that they measured the landscape structure at the three neighborhood levels: 750 ft (229 m), 1500 ft (457 m), 1500 ft (457 m) radius from the location of the respondent. However, the observations were done separately, not spontaneously. Schlöpfer et al. (2015) designated communities of four types—urban centers, suburban, high-income, and peri-urban—and ran two-level models to estimate the economic impact of environmental amenities in apartment rental price in Switzerland. However, landscape inventories in this study were also limited to precisely measure the impact of landscape structure characteristics on residential housing rents. A radius of 300 m-buffer was used to construct the neighborhood variables.

2.4. Literature gap and research questions

In light of those findings from the literature, this study addresses several gaps. First, while the role of landscape patterns on housing premiums has frequently been studied, these examinations have been limited to the presence, proximity (e.g., visibility and distance), and ratio of them in neighborhoods. Second, the measures of neighborhood satisfaction are diverse in studies in the past, but housing price can bring benefits since the sample size is relatively larger than survey collected data and is an effective proxy by which to measure objective and subjective evaluation of residents on landscape. Third, the definition of neighborhoods has often been limited to buffer-drawn areas, which differ from real neighborhoods as perceived by the general public. Finally, inappropriate statistical operations have often been conducted by assigning contextual conditions of neighborhoods to nested parcels while using neighborhoods with predefined boundaries and OLS. To fill these gaps, this study specifically focuses on (1) whether the composition and configuration of landscape in neighborhoods regarding size, shape, isolation, and connectivity can exert influence over housing prices and (2) whether MLM explains the impact of neighborhood landscape patterns on housing prices that is theoretically and statistically appropriate by comparing the results of OLS that frequently appeared in the previous empirical studies.

3. Research design

3.1. Study area and units of analysis

The study area is Austin, Texas (see Fig. 1.). Information on housing structures and characteristics was collected at the parcel level, and data on landscape patterns and other neighborhood conditions were gathered from residential subdivisions that shared the same name. Subdivisions with identical names were assumed to most aptly represent residential “neighborhoods,” since they tend to possess similar characteristics such as age of development, patterns of urban form, income levels, and life-cycles (Blake and Arreola, 1996). The homogeneity of development age, housing price, lot size, and housing size was evaluated through the coefficient of variation (CV), which demonstrated that most of the subdivisions used in this study were relatively homogeneous. The CV serves as a normalized measure of data dispersion of the mean, calculated as the ratio of the standard deviation to the mean (Osborne and Overbay, 2004). A CV value of less than one

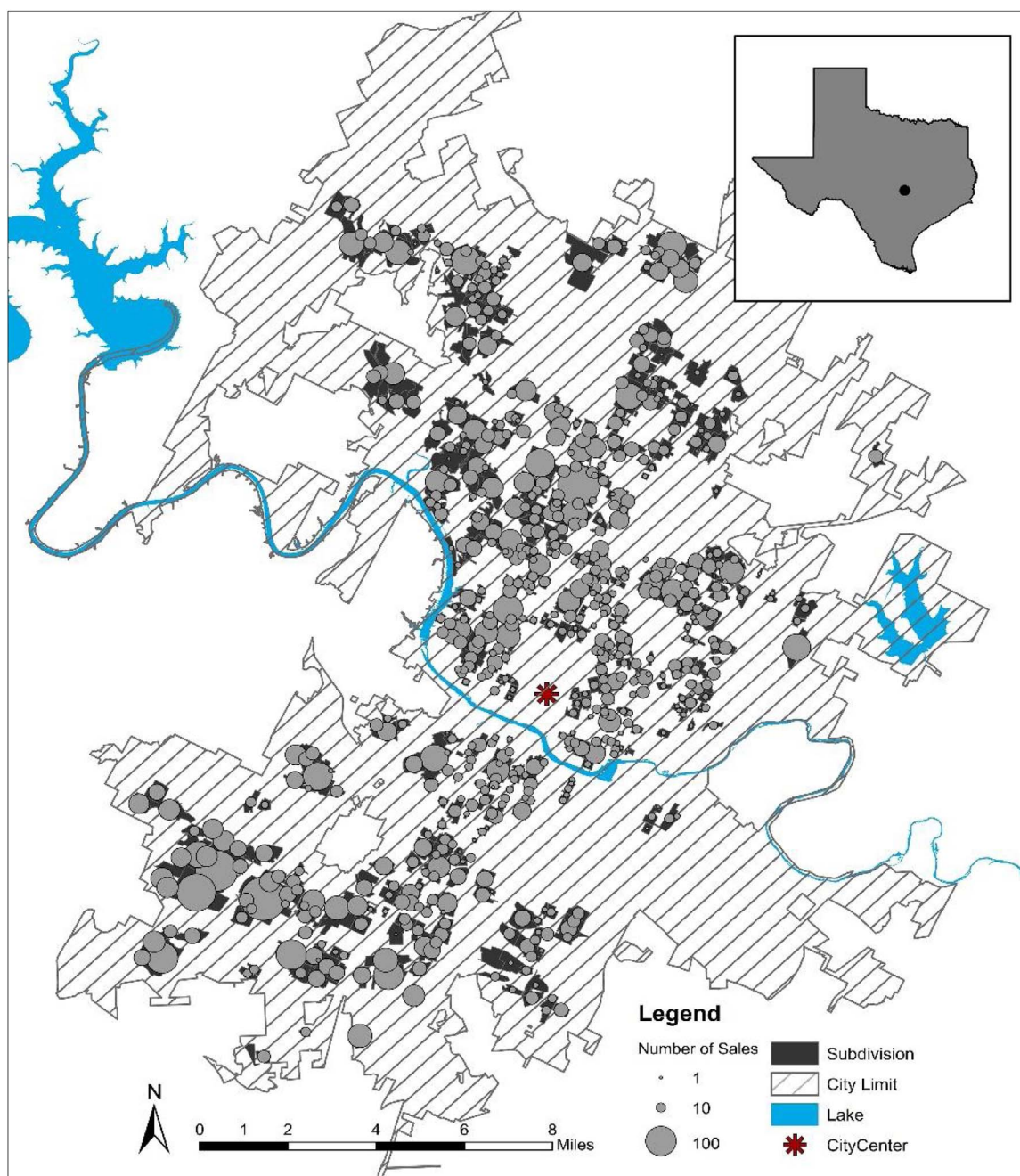


Fig. 1. Study area: Austin, TX and total numbers of sales in each subdivision.

generally indicates that the units of interest are relatively homogeneous (Faria Filho et al., 2010). There were a few subdivisions with CV values greater than one: zero subdivisions in development age, nine in lot size, six in housing size, and four in appraised housing value out of 649 subdivisions. This, however, did not impact the final results. Subdivisions with less than 30 housing units were excluded from the study due to their extremely small size. On average, subdivisions in this study had 49 acres with 142 single-family houses (337 people with an average household size of 2.37 in 2010).

3.2. Data and measurements

Even though landscapes can include a wide variety of aggregated landforms (Forman and Godron, 1981), this study intentionally highlighted only tree cover—one of the major components of landscape in neighborhoods. To measure these landscape patterns, a tree cover map

was produced according to the steps suggested by Behee (2012), since the National Land Cover Database (NLCD) has a coarse resolution of 30-m. One-meter infrared color high-resolution digital ortho quadrangle (DOQs) images, updated as of 2010, from the Texas Natural Resources Information System (TNRIS) were incorporated into normalized difference vegetation index (NDVI) layers. Texture analyses were added above the four bands provided (near infrared, red, green, and blue) in the DOQs imagery. A texture analysis was included to distinguish the texture and shapes between trees and grass, helping to increase accuracy when separating forested areas and evenly illuminated low grass areas (Zhang, 2001). A seven-by-seven pixel size was chosen to identify mature tree crowns—a radius of seven pixels was generally interpreted as one mature tree crown in a one-meter resolution image. An ISO cluster unsupervised classification was conducted with six layers: band 1 (blue), band 2 (green), band 3 (red), band 4 (Near-IR), texture, and created NDVI. Seventy classes of layers were created and

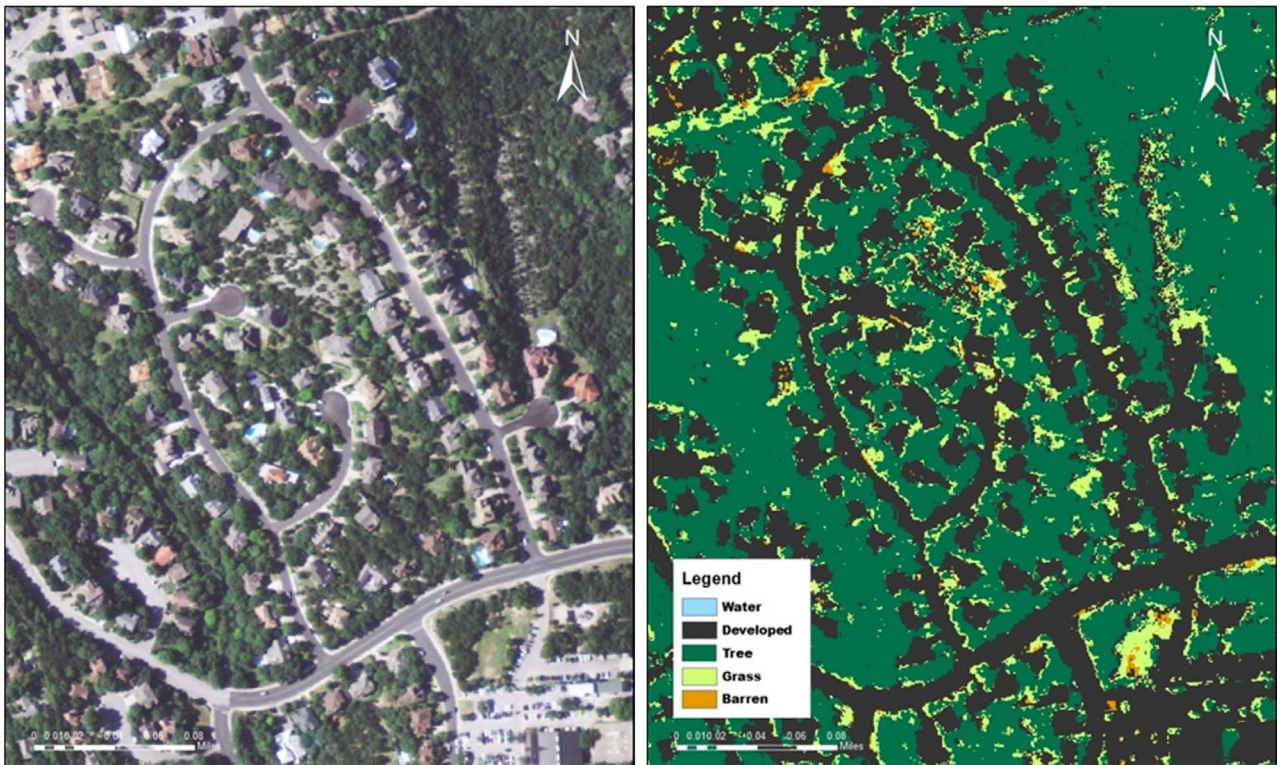


Fig. 2. Created land cover data with five classifications: water, developed, tree, grass, and barren.

organized into five classes: water, developed area, tree, grass, and dry grass/barren (see Fig. 2.).

Five landscape ecology metrics identified the “size,” “edge,” “shape,” “isolation,” and “connectivity” of landscape patterns. The size of tree cover was measured by the percentage of tree cover (PLAND) and patch density (PD); the edge by edge density (ED); the shape by contiguity index (CONTIG) and mean shape index (SHAPE); isolation by proximity (PROX) and Euclidean nearest distance (ENN); and connectivity by connectedness (CONNECT) and Cohesion (COHE). Detailed measurements are provided in Table 1. Only PLAND, ED, SHAPE, and CONNECT were added to the final model because (1) these metrics are well-suited for intuitive interpretation and (2) there was a high multi-collinearity between nine variables. High values of PLAND, ED, SHAPE, and CONNECT indicated the presence of larger, finer, more complex, and more connected tree cover patterns within subdivisions, respectively.

Variables with potential correlations to housing prices at the subdivision level, such as school quality, were measured using the ratio of the exit-level passing rate of the 11th grade Standard Accountability Indicator (SAI) of secondary schools to the stated SAI in 2010 for each assignment school district, as well as the amenity of proximity to lakes and major roads (proximity to lakes serves as a critical determinant of housing premiums). At the parcel level, components of housing structure (typically referred to as determinants of housing prices in the literature) were measured, such as the size of the living area (LIVING) and lot (LOT); the number of bedrooms (BED) and bathrooms (BATH); the number of floors (STORY); the presence of a fireplace (FIRE), garage (GARAGE), or pool (POOL); and the year built (YEAR) (Anthon et al., 2005; Conway et al., 2010; Donovan et al., 2011; Li et al., 2015; Schlöpfer et al., 2015; Wen et al., 2015). Only LIVING, LOT, YEAR, STORY, FIRE, GARAGE, and POOL were eventually used in the final model due to issues of multi-collinearity.

A total of 9958 single-family housing transactions containing the above housing structure characteristics between 2009 and 2011 were obtained from the Austin Board of REALTORS®. A total of 224 proper-

ties with lot sizes of “zero” were excluded beforehand due to miscoding, even though they otherwise had complete information. All other data sets were retrieved from the city of Austin, the Travis Central Appraisal District (TCAD), the Texas Education Agency, and TNRIS. ArcGIS 10.2 and FRAGTSTATS 4.2 (McGarigal et al., 2012)—computer software designed to calculate landscape patterns—were used for data collection and measurement. Housing price is log transformed in the analyses.

3.3. Models

Four MLM models—“ANOVA”, “random-coefficient”, “means as outcomes”, and “intercepts and slopes as outcomes”—were used to analyze the cross effects of landscape patterns at parcel and subdivision levels on housing sales prices. An HLM 7 software package (Raudenbush et al., 2011) was used for these analyses. The ANOVA model in Eq. (3) served as a preliminary step for cross-level analysis, testing the degree of significant variance in housing price between subdivisions. Such variation of housing prices across subdivisions necessitates the use of an MLM. The ANOVA has no independent variable at either the parcel or the subdivision level.

$$\begin{aligned} \ln Price_{ij} &= \beta_{0j} + \gamma_{ij} \\ \beta_{0j} &= \gamma_{00} + \mu_{0j} \end{aligned} \quad (3)$$

where $\ln Price_{ij}$ denotes the log transformed price of the i th parcel in the j th subdivision. β_{0j} is a parcel level coefficient; γ_{ij} represents a parcel-level random effect; and μ_{0j} indicates a subdivision-level random effect.

The random coefficient model—which included only explanatory variables at the parcel level in addition to the ANOVA shown in Eq. (4)—was used to identify the impact of the independent variables at the parcel level and to compare the degree to which conditions at the subdivision level accounted for variance across subdivisions. This was achieved by calculating the reduced variance upon adding subdivision level predictors at intercepts and slopes as outcomes. LIVING and YEAR were treated as random effects, while the others were deemed to be fixed for two reasons: (1) the slopes of LIVING and YEAR in housing

Table 1
Variables of interest and descriptive statistics.

Variable	Measurement	Source; Analytical tools	Range	Mean	S.D.	V*
<i>Subdivision level (N = 649)</i> Size/Edge	Percentage of landscape	PLAND = The percentage share of the tree cover, units in percentages	[5.13, 84.86]	49.51	11.82	IV
Shape	Patch density	PD = The number of patches per unit area	[610.84, 8072.37]	2465.65	1109.95	–
	Edge density	ED = Total length of edge of a particular class per unit	[507.52, 2239.35]	1676.54	216.03	IV
	Shape index	Normalized ratio of patch perimeter to area	[1.20, 1.56]	1.36	0.06	IV
		SHAPE = $\frac{p_{ij}}{\max p_{ij}}$				
Contiguity index		p_{ij} = perimeter of patch i in class j in terms of number of cell surfaces.				
		Average contiguity value for the cells in a corresponding patch	[0.13–0.49]	0.25	0.04	–
		CONTIG = $\left[\frac{\sum_{j=1}^v q_{ij}}{q_{ij}} \right] - 1$				
		$v - 1$				
Isolation		c_{ijr} = contiguity value for pixel r in patch ij				
		a_{ij} = area of patch ij in terms of number of c				
		v = sum of the values in a 3-by-3 cell template				
		Sum of all patches of the corresponding patch type whose edges are within an 0.5-mile radius of the focal patch, of each patch size divided by the square of its distance from the focal patch	[1.18, 82978.95]	4665.75	8212.93	–
Proximity		PROX = $\sum_{i=1}^n \frac{S_i}{z_i}$				
		S_i = area				
		z_i = edge to edge distance from patch i to its nearest neighbor indexed patch identified within the buffer				
		ENN = $\frac{\sum_{j=1}^n \sum_{i=1}^n h_{ij}}{N}$				
Euclidean nearest distance		h_{ij} = the edge-to-edge distance from patch i in class j to the nearest neighboring patch of the same class				
		N = the number of patches in the landscape that have nearest neighbors	[2.21, 6.30]	2.84	0.32	–
		Number of functional joins between patches of the same type within 0.5 miles divided by the total number of possible joins between all corresponding patches multiplied by 100				
		CONNECT = $\left[\frac{\sum_{j=k}^n c_{ijk}}{n_i(n_i - 1)} \right] \times 100$				
Connectivity		c_{ijk} = joining between patch j and k (0 = unjoined, 1 = joined) of the corresponding patch type (i), based on a user specified threshold distance.				
		n_i = number of patches in the landscape of the corresponding patch type				
		Proportional to the area-weighted mean perimeter-area ratio divided by the area-weighted mean patch shape index				
		COHESION = $\left(1 - \frac{\sum p}{\sum p \cdot \bar{a}} \right) \left(1 - \frac{1}{\sqrt{N}} \right)^{-1}$				
Cohesion		p = patch perimeter				
		a = patch area				
		N = the number of pixels on the map				
		The presence of a lake within a 500 ft. buffer from the edge of a subdivision (if yes = 1, no = 0)				
Other conditions	Lake	The presence of major arterials and a metro rail within a 500 ft. buffer from the edge of a subdivision (if yes = 1, no = 0)	[0.1]	0.37	0.48	IV
	Traffic	The ratio of 11th grade Standard Accountability Indicator (SAI) of secondary schools' exit-level passing rate to the stated SAI in 2010 for each assignment school district	[0.1]	0.69	0.46	IV
	School quality		[44.00, 95.00]	69.18	14.54	IV
Parcel level (N = 9958)	Sales price	House sales price (2009–2011), log transformed in the model	[29000, 6300000]	291159.91	220962.49	DV
	Years built	Year a house was built	[1877, 2012]	1972.67	20.44	IV

(continued on next page)

Table 1 (continued)

Variable	Measurement	Source; Analytical tools	Range	Mean	S.D.	V*
Story	The number of house floors		[1,3]	1.30	0.49	IV
Fire	Existence of a fireplace (if yes = 1, no = 0)		[0,1]	0.67	0.58	IV
Garage	The number of garages		[0,6]	1.41	0.89	IV
Pool	Existence of a pool (if yes = 1, no = 0)		[0,1]	0.07	0.26	IV
Living area	Total living area in square feet		[100.00, 9615.00]	1847.95	784.88	IV
Lot size	Lot area in square feet		[2047.32, 123884.64]	9473.50	5256.79	IV
Bathroom	Number of full and half baths. A half-bath is counted as 0.5.		[0,7]	2.144	0.73	-
Bedroom	Number of bedrooms		[0,11]	3.25	0.74	-

V* refers to whether each variable was included in the final models (DV: dependent variable, IV: independent variable -: not included).

prices varied significantly across subdivisions, while the slopes of the others were similar, and (2) there was no back-up theory indicating that other housing characteristics should differ across subdivisions.

$$lPrice_{ij} = \beta_{0j} + \beta_{1j}(YEAR) + \beta_{2j}(STORY) + \beta_{3j}(FIRE) + \beta_{4j}(GARAGE) \\ + \beta_{5j}(POOL) + \beta_{6j}(LIVING) + \beta_{7j}(LOT) + \gamma_{ij} \\ \beta_{qj} = \gamma_{q0} + \mu_{qj} \quad \text{for } q = 0, 1, 6 \\ \beta_{qj} = \gamma_{q0} \quad \text{for } q = 2, 3, 4, 5, \text{ and } 7 \quad (4)$$

where all parcel level predictors are group-mean centered.

The mean as outcome model—which included only variables at the subdivision level on top of the ANOVA shown in Eq. (5)—was conducted to identify the influence of subdivision conditions alone, independent from parcel characteristics.

$$lPrice_{ij} = \beta_{0j} + \gamma_{ij} \\ \beta_{0j} = \gamma_{00} + \gamma_{q1}(PLAND) + \gamma_{q1}(ED) + \gamma_{q1}(SHAPE) + \gamma_{q1}(CONNECT), \\ + \gamma_{q1}(LAKE) + \gamma_{q1}(TRAFFIC) + \gamma_{q1}(SCHOOL) + \mu_{0j} \quad (5)$$

where all subdivision level predictors are grand-mean centered.

Finally, Eq. (6) was run under the model of intercepts and slopes as outcomes, accounting for all predictors at both levels to directly test the joint effects of parcel and subdivision conditions. All parcel predictors were group-mean centered—that is, centered by the mean of each subdivision. Subdivision predictors were also group-mean centered to compare differences across subdivisions.

$$lPrice_{ij} = \beta_{0j} + \beta_{1j}(YEAR) + \beta_{2j}(STORY) \\ + \beta_{3j}(FIRE) + \beta_{4j}(GARAGE) \\ + \beta_{5j}(POOL) + \beta_{6j}(LIVING) + \beta_{7j}(LOT) \\ + \gamma_{ij} \\ \beta_{0j} = \gamma_{00} + \gamma_{q1}(PLAND) + \gamma_{q1}(ED) + \gamma_{q1}(SHAPE) \quad \text{for } q = 0, 1, 6 \\ + \gamma_{q1}(CONNECT) + \gamma_{q1}(LAKE) \\ + \gamma_{q1}(TRAFFIC) + \gamma_{q1}(SCHOOL) + \mu_{0j} \\ \beta_{qj} = \gamma_{q0} \quad \text{for } q = 2, 3, 4, 5, \text{ and } 7 \quad (6)$$

where all parcel level predictors are group-mean centered and subdivision level predictors are grand-mean centered.

4. Results

4.1. MLM models

The interclass correlation coefficient (ICC), calculated based on the outcomes of the ANOVA test, indicated that 83.5% ($ICC = \tau_{00 \text{ ANOVA}} / [\tau_{00 \text{ ANOVA}} + \sigma^2_{\text{ANOVA}}] = 0.353 / [0.353 + 0.070] = 0.835$) of the variance in mean sales prices (logged) occurred between subdivisions and that 16.5% ($= 100 - 83.5 = 16.5\%$) occurred within subdivisions. The χ^2 test value of 40,932.32 ($p < 0.001$) also suggested that the variation among subdivisions in mean sales differed significantly from zero. Thus, the difference in mean sales price seems to arise not at the parcel level but at the subdivision level, such that a multi-level analysis would yield unbiased results and therefore proves to be more effective than a single-level model.

The approximate R^2 is the MLM measure that explains variance based on the reduction in variance components when adding predictors (LaHuis et al., 2014). This differs from the total R^2 in OLS models in that the explained variance is calculated at both the parcel and subdivision levels, as suggested by Bryk and Raudenbush (1992). When the random coefficient and ANOVA models were compared, the parcel-level predictors alone explained 48.6% (within explained variance

Table 2
Estimated random effects..

Models	Parameter	Std. dev.	Variance component	D.F.	χ^2	p-value
ANOVA	Intercept (τ_{00})	0.594	0.353	648	40932.320	< 0.001
	Level-1 (σ^2)	0.264	0.070			
Means as outcomes	Intercept (τ_{00})	0.421	0.178	641	19419.29	< 0.001
	Level-1 (σ^2)	0.265	0.070			
Random coefficient	Intercept (τ_{00})	0.601	0.361	549	68380.700	< 0.001
	Level-1 (σ^2)	0.189	0.036			
Intercepts & slopes as outcomes	Intercept (τ_{00})	0.430	0.185	542	32046.988	< 0.001
	Level-1 (σ^2)	0.189	0.036			

τ_{00} : true subdivision means around the grand mean and σ^2 : true parcel means around the group mean.

$(R^{2*}) = [\sigma_{ANOVA}^2 - \sigma_{Random\ Coefficient}^2] / \sigma_{ANOVA}^2 = [(0.070 - 0.036) / 0.070] = 0.486$ of the variance in mean sales prices. Total explained variance occurring between subdivisions is 49.6% ($R^{2*} = [\tau_{00\ ANOVA} - \tau_{00\ Means\ as\ outcomes}] / \tau_{00\ ANOVA} = [(0.353 - 0.178) / 0.353] = 0.496$). This indicates that turnover explained by the subdivision level characteristics has relatively similar explanatory power of parcel predictors, recalling the 48.6% explained variance occurred within subdivisions. Adding predictors of subdivisions on top of parcel predictors—with a comparison between the random coefficient model and the model of intercepts and slopes as outcomes (the fitted model)—reduces the variation in mean sales prices by 48.8% (between explained variance (R^{2*}) = $[\tau_{00\ Random-Coefficient} - \tau_{00\ Fitted}] / \tau_{00\ Random-Coefficient} = [0.361 - 0.185] / 0.361 = 0.488$). The values of τ_{00} , σ^2 , and χ^2 in each model are presented in Table 2.

The outcomes of the MLM models can be explained in a way similar to that of the OLS outcomes. As described in Table 3, if a house is newly built ($r = 0.004$, $p < 0.001$) on a large lot ($r = 0.000$, $p < 0.001$) with a large living space ($r = 0.000$, $p < 0.001$), garages ($r = 0.030$, $p < 0.001$) and a pool ($r = 0.060$, $p < 0.001$), its price is likely to be high. However, the presence of multiple stories tends to reduce the price ($r = -0.016$, $p = 0.009$). At the subdivision level, the presence of a large proportion of tree cover ($r = 0.020$, $p < 0.001$) contributed to increased housing premiums. Average sales prices increased by about 2% for every 1 unit increase in tree cover when all other factors remained constant. A house valued at \$220,962—the mean sales price of our study sample—would see an average increase of \$4419 if a subdivision were to add 1% more tree cover. Other landscape patterns such as edge density, shape, and connectivity were found to be statistically insignificant factors.

Other conditions such as proximity to lakes ($r = 0.0556$, $p = 0.157$) and major roads ($r = -0.0078$, $p = 0.837$) were insignificant, but inclusion within a quality school district ($r = 0.020$, $p < 0.001$) increased the housing price. The results suggest that a 1 point increase in the ratio of SAI of a school to a state would lead to an increase in housing price of 2%, which translates to approximately \$4419 for a house valued at \$220,962.

The outcomes of the MLM also indicate that some associations exist between parcel and subdivision conditions. Edge density of tree cover alone does not produce an impact on housing prices, but the slope/rate of change between the year built and the mean sales price can decrease by 0.001% for every one-point increase in edge density. To facilitate understanding, the interaction term was explained in terms of low (the mean minus one standard deviation) and high (the mean plus one standard deviation) conditions. For every one-year increase in the years that a subdivision was built, the mean sales price was found to decrease by 1.5% if the subdivision has a high edge density ($= 0.0039 + (-0.00001 \times 1892.6) = -0.015$); low edge density corresponded to a decrease in housing premiums by 1.07% ($= 0.003914 + (-0.00001 \times 1460.5) = -0.0107$); 1892.6 is one standard deviation above from the mean of edge density and 1460.5 is one standard deviation below from the mean edge density. This

indicates that the impact of the year in which a subdivision was built can depreciate due to landscape patterns, especially edge density, though this may not be substantial.

Similarly, for subdivisions with high quality schools, every one square-foot increase in living area corresponds to a mean sales price increase of 0.014% ($= 0.000305 + (-0.000002 \times 83.72) = 0.000138$), meaning a \$30.9 increase for a house valued at \$220,962. For subdivisions with low quality schools, the mean sales price would increase by 0.020% ($= 0.000305 + (-0.000002 \times 54.64) = 0.000196$), or about \$44.2 for the same house. When the house's living space and area's school quality are considered together, the school quality factor produced an impact in the opposite direction, compared to the case of considering school quality independently.

4.2. Comparing results from MLM and OLS

As shown in Table 3, two OLS models were compared with the MLM. The first OLS has been centered by its group- and grand-mean to facilitate close comparison to the MLM models that is also centered. The second OLS is not centered and is therefore similar to the traditional OLS. The centered and un-centered OLS models show different results from those of previous studies and of the MLM in this study. Un-centered OLS reported that years built, fire place, and lot size had negative impacts on housing premiums, despite being frequently mentioned as positive factors in previous studies (Li and Brown, 1980; Li et al., 2015; Shin et al., 2011). Meanwhile, centered OLS supported the consistent results with the findings from the previous studies; years built, fire, garage pool, living area and lot size yield positive influences in increasing housing price. The percentages of landscape and shape produced statistically significant impact across three models: a positive impact for percentage of tree ($OLS_{centered} = 0.02$ and $OLS_{un-centered} = 0.01$) and edge density ($OLS_{centered} = 0.00$ and $OLS_{un-centered} = 0.00$) and a negative impact by tree shape ($OLS_{centered} = -0.71$ and $OLS_{un-centered} = -0.70$). However, MLM reported that only the percentage of tree cover contributed to the increased housing price. Other subdivision conditions such as lake, traffic, and school quality were also explained. The quality of high schools was considered to be a favorable condition in all three models, whereas the presence of lakes was positive in the centered model and negative in the un-centered OLS model.

5. Discussion

5.1. Findings from the MLM model

This study found that the most of the housing structures were found to be statistically significant, with factors aligning with those found in the previous literature (Jim and Chen, 2009; Sander et al., 2010; Troy and Grove, 2008), except for that of the fireplace ($r = 0.009$, $p = 0.154$). While fireplaces are typically perceived as amenities (Li

Table 3
Results of MLM and OLS (centered and un-centered).

Characteristics	MLM			OLS (centered)			OLS (un-centered)		
	Coefficient	Stand. Error	p-value	Coefficient	Stand. error	p-value	Coefficient	Stand. error	p-value
<i>Subdivision level (N subdivision = 649)</i>									
Constant	12.3543 [*]	0.0170	0.000	0.0193 [*]	0.0004	0.000	0.0137 [*]	0.0004	0.000
Landscape pattern									
Percentage	0.0197 [*]	0.0017	0.000	0.0001 [*]	0.0000	0.000	0.0003 [*]	0.0000	0.000
Edge density	0.00001	0.0001	0.481	−0.7134 [*]	0.0823	0.000	−0.7010 [*]	0.0692	0.000
Shape	0.3590	0.3062	0.241	0.0008	0.0005	0.114	0.0013 [*]	0.0004	0.002
Connectivity	−0.00001	0.0020	0.994	0.0211 [*]	0.0085	0.013	−0.0141 [*]	0.0071	0.047
Lake	0.0556	0.0393	0.157	0.0171	0.0094	0.070	0.0417 [*]	0.0079	0.000
Traffic	−0.0078	0.0379	0.837	0.0158 [*]	0.0003	0.000	0.0095 [*]	0.0003	0.000
School Quality	0.0197 [*]	0.0013	0.000	12.2912 [*]	0.0106	0.000	17.4669 [*]	0.4643	0.000
<i>Parcel level (N parcel = 9958)</i>									
Year built	0.0039 [*]	0.0004	0.000	0.0042 [*]	0.0004	0.000	−0.0035 [*]	0.0002	0.000
× Percentage	−0.0001	0.0000	0.084						
× Edge density	−0.00001 [*]	0.0000	0.000						
× Shape	0.0001	0.0060	0.992						
× Connectivity	0.00005	0.0007	0.864						
× Lake	−0.0001	0.0008	0.228						
× Traffic	−0.0010	0.0000	0.240						
× School quality	−0.00003	0.0000	0.238						
Story	−0.0157 [*]	0.0063	0.009	−0.0094	0.0115	0.415	−0.0021	0.0087	0.807
Fire	0.0094	0.0066	0.154	0.1107 [*]	0.0082	0.000	−0.0445 [*]	0.0077	0.000
Garage	0.0304 [*]	0.0051	0.000	0.0273 [*]	0.0070	0.000	−0.0076	0.0050	0.127
Pool	0.0603 [*]	0.0072	0.000	0.1516 [*]	0.0165	0.000	0.0023	0.0139	0.870
Living area	0.0003 [*]	0.0000	0.000	0.0002 [*]	0.0000	0.000	0.0005 [*]	0.0000	0.000
× Percentage	0.0000	0.0000	0.917						
× Edge density	0.0000	0.0000	0.571						
× Shape	−0.0002	0.0002	0.265						
× Connectivity	0.000004 [*]	0.0000	0.005						
× Lake	0.000004	0.0000	0.823						
× Traffic	0.00002	0.0000	0.439						
× School quality	−0.000002 [*]	0.0000	0.013						
Lot size	0.000004 [*]	0.0000	0.000	0.0003 [*]	0.0001	0.000	−0.000003 [*]	0.000001	0.000
Between R ² = 48.8%				Adj. R ² = 52.5%			Adj. R ² = 67.4%		

* $p < 0.05$.

and Brown, 1980), they were not statistically significant in this particular study, since the city of Austin boasts relatively moderate winter weather—with an average high temperature of 69 °F. Heating from a fireplace was not an relevant determinant of housing price in this location. At the subdivision level, at variance from the assumption of this study, homebuyers seemed to prioritize the composition of tree cover but not necessarily their spatial configuration. This finding is partially supported by previous empirical research that emphasized the value of larger green spaces. Forman and Godron (1981) and Dramstad et al. (1996) argued that larger open spaces allow neighborhoods to be more livable and valuable to their residents. As mentioned in the literature review section, in theory having a large amount of trees in an area meets basic human psychological needs (1) by providing a clear view to prospect and safer places for refuge as described in the prospect-refuge theory (Appleton, 1996); (2) offering restoration from attentional fatigue as described in the psycho-evolutionary model (Ulrich et al., 1991); and (3) influencing the cognitive process to determine people's preference as indicated in informational theory (Kaplan, 1975). For such reasons, people have shown a consistently higher preference of green spaces and natural environments (Hartig and Staats, 2006). Still, it is unclear as to why other spatial patterns such as shape, isolation, and connectivity are not statistically significant. One source of evidence could be the size of subdivisions used in this study. The subdivisions in this study may not be large enough to discern significantly different landscape spatial patterns within their boundaries. Geoghegan et al. (1997) observed spatial patterns of landscape in terms of diversity and fragmentation, as well as the percentage of forest, agriculture, and urban open space in the Patuxent Watershed in central Maryland, and their findings partially support this argument. Their study found that landscape diversity, measured within a 0.1-km buffer

was not statistically significant at the 10% level but was in fact significant at a 1-km buffer. In terms of size, a 0.1-km buffer covers about 78 acres of land, slightly larger than the subdivisions in this study (49 acres), while a 1-km buffer contains 776 acres of land. The percentage of open space impacted housing premiums at both spatial levels, although the ratio of open space functioned as a positive factor in the 0.1-km buffer area and a negative factor in the 1-km buffer. This implies that residents were more sensitive to the total amount of landscape than the spatial arrangement of the landscape when observing micro-scale neighborhoods. Another possibility is that the total number of trees was the most visible factor for home buyers who were less familiar with details about subdivisions and that they may have needed additional time to ascertain spatial details of landscape in these neighborhoods.

In addition, the school quality in relation to determining housing price was found to be critical, which aligns with previous literature that has elaborated on the positive impact of school quality on increased housing prices (Downes and Zabel, 2002; Gibbons and Machin, 2005; Leech and Campos, 2003). Even though the impact of other conditions such as proximity to lakes and major roads had been often mentioned in previous literature (Li et al., 2015; Wen et al., 2015), this impact was not always considered to play a significant role on housing prices. For instance, Song and Knaap (2004) found that distance to major roads functioned as a positive and significant factor, while Brown and Pollakowski (1977) argued that distance to the nearest lake served as an insignificant determinant of housing premiums. It is possible that the visibility of a lake proves more critical than a house's simple proximity to a lake (Lansford and Jones, 1995; Seiler et al., 2001). Proximity was found not to be a significant factor in this study, but at the same time, proximity to major roads—i.e., within 500 ft in this study—was found

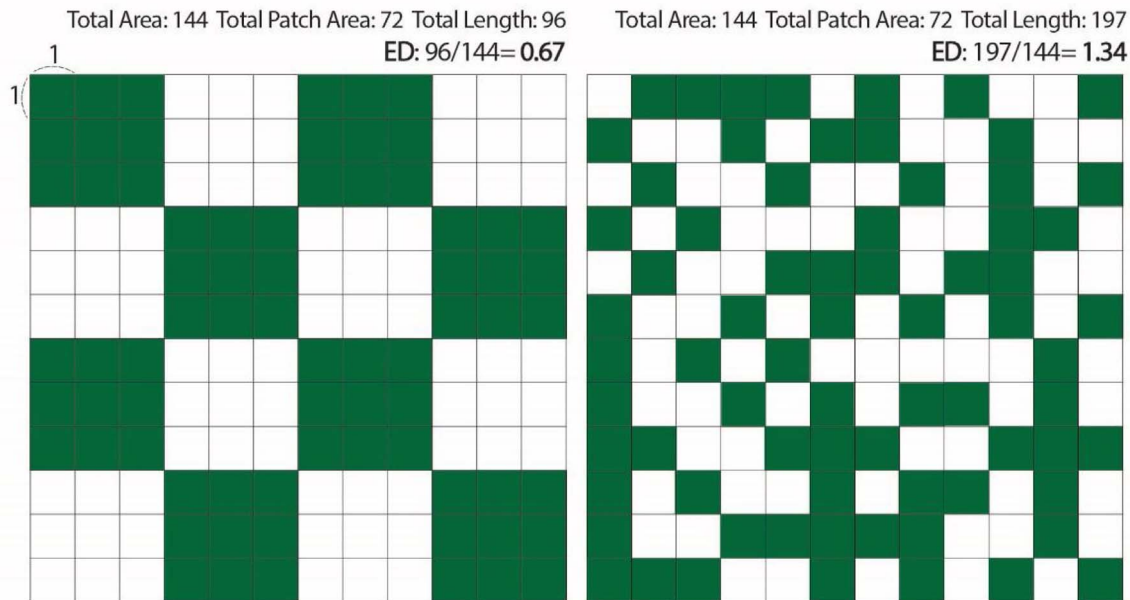


Fig. 3. Comparison of different landscape patterns: high ED refers to finer-grained landscape.

to be an undesirable condition due to increased traffic, congestion, and noise (Bin, 2004).

The study also found some combinational association between parcel and subdivision characteristics: years of a house since built and edge density and the size of living area and tree connectivity and school quality. There are no concrete supporting theories or scholarly empirical studies explaining why these combinations could be seen as favorable or unfavorable conditions. Typically, trees or forests transition from fine-grained to coarsening conditions over time (van Noordwijk et al., 2014). An old house is often accompanied by an assumption that the first development of the subdivision occurred a long time ago, such that a coarse-grained landscape—“coarse-grained” indicating a lower value of edge density, as shown in Fig. 3—would be expected. Thus, if an old house is surrounded by fine-grained tree cover, this might mean that the landscape not healthy, implying that the trees might be immature or poorly managed in the subdivision. This also corresponds with the previous sense of place concepts that not just the types of landscape but also nature and physical characteristics affect people's attachment to a specific place (Adevi and Grahn, 2012).

There is also no generalized argument to explain why the association between school quality and living area is unfavorable condition. In Austin, subdivisions with large living areas in sold property are likely to have proximity to the city center, as shown in Fig. 4. It is possible that residents living in the city center tend not to have as many children, due to being young professionals or empty nesters. As a result, finding a good school district may not be a particularly high priority for them. In fact, these residents might actually prefer to avoid subdivisions with good schools that tend to attract families with children, which could be a negative condition for them.

Finally, the study also compared three different models to identify using the MLM model could be a statistically and theoretically appropriate approach. The comparison of three models implies that using a different model could lead to dissimilar findings. Although it is hard to say which model reports the “best” outcomes that reflect reality, using traditional OLS does not seem to be the best approach, in that factors (i.e., lake, years built, fireplace, and lot size) found as negative in the un-centered OLS were believed to be positive determinants on housing premium in many previous studies. Centered-OLS models generated relatively consistent results with the previous research and were useful to interpret variables with different units of scale, but centering has not commonly used in OLS regression. Also, it does not

automatically create interaction terms between the parcel and subdivision levels. To create interaction terms in OLS between different scales, the theory should be supported.

6. Conclusions

According to both MLM and OLS models, the composition of landscapes, especially tree covers in a subdivision, did indeed have an impact on housing prices. However, other spatial patterns related to landscape such as density, shape, and connectivity, did not show statistically significant effects. Although the OLS reported that a fine-grained landscape pattern may be a potential threat to housing premiums, the MLM indicated that this condition might present an issue only when it is considered together with housing age; an older house surrounded by fine-grained landscape could function as a possible negative factor on housing premiums. Another contribution of this study comes from its use of residential subdivisions to closely match the meaning of neighborhoods, as well as its use of MLM to explain the characteristics of multi-level units in statistically and conceptually appropriate ways. The results from the OLS models, more common in previous studies, showed different results than those of the MLM models. In addition, the OLS reported results that contradicted those of previous studies for some features (e.g., years built, fire place, and lot size), especially with an un-centered model. Therefore, this study encourages the use of an MLM approach for future studies that examine multi-level units with a hierarchical structure.

These results of this study can contribute to discussions on the association between spatial patterns of landscape and housing premiums, but some points remain that future studies should consider. First, the socio-economic characteristics of each subdivision went unaddressed in this study due to the lack of available data and due to the selective focus of this study on landscape patterns. Future studies would be advised to include socio-economic data retrieved from a census. To make this possible, three-level modeling—parcels, subdivisions, and census tracts from lower to higher order—could be utilized for optimal conceptual and operational justification. Second, this study examined landscape patterns and housing prices in one city over a relatively short period of time. Austin is located in the Southern region of the U.S. and its particular lifestyle and affinity for certain design elements might not align with the preferences of other cities. To generalize these findings, future studies should account for a greater

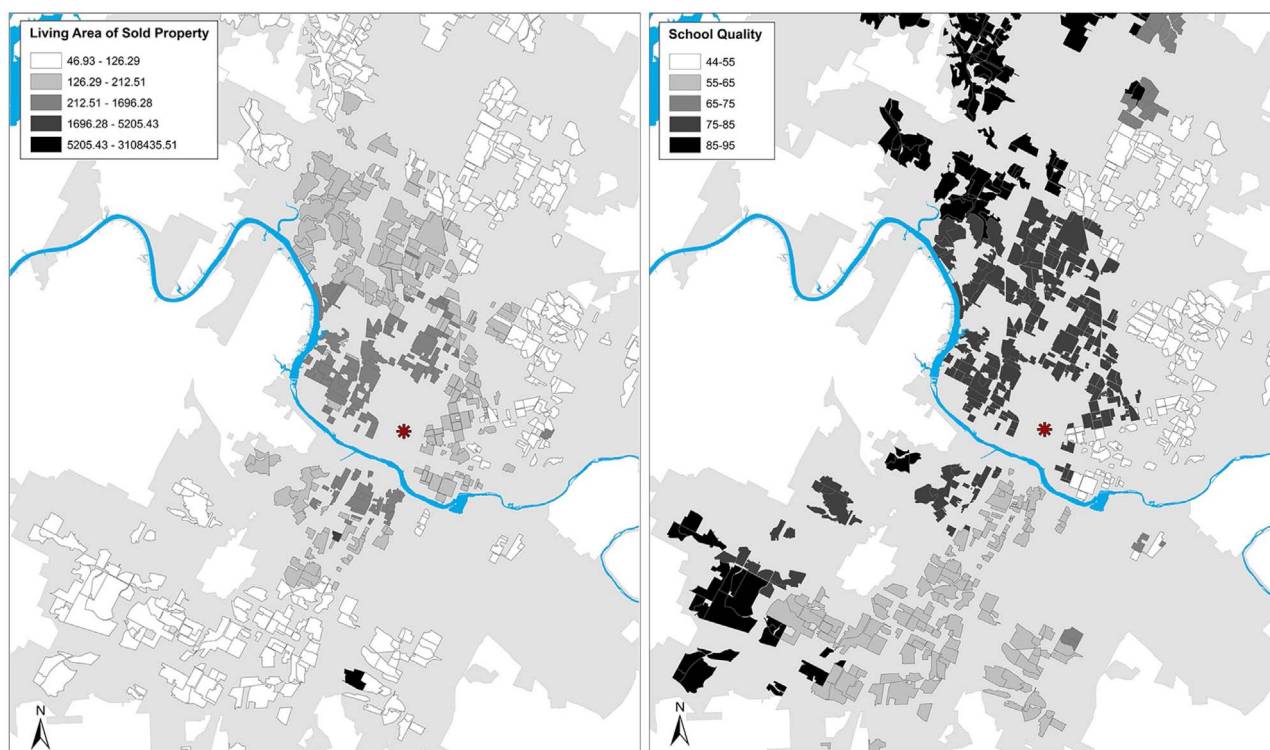


Fig. 4. Distribution of mean living areas and school quality in Austin neighborhoods.

number of cities of various sizes, phases, and regions. Due to the lack of data availability, only single-family houses were investigated in this study. Thus, the outcome of this study might not comprehensively reflect the housing prices and conditions of other types of housing.

Considering these possibilities, the outcomes of this study suggest that discussions about detailed spatial configurations of landscape at the micro-level of neighborhoods might not be as productive as they would be in macro-level areas such as cities, counties, or regions. The discussion would certainly benefit from additional empirical tests on how these landscape patterns affect housing premiums at macro-level communities. Green infrastructure initiatives, programs, and plans should be developed based on the spatial structure of landscape patterns.

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